

(a) Find an equation of Π_1 , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$. [2]

[illegible]

(b) Find an equation of Π_2 , giving your answer in the form $ax + by + cz = d$. [4]

[illegible]

- (c) Find the acute angle between Π_1 and Π_2 . [5]

[illegible]

- (d) The point Q is such that $\overrightarrow{OQ} = -5\overrightarrow{OP}$.

Find the position vector of the foot of the perpendicular from the point Q to Π_2 . [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.